# Notes: Lazear (JPE, 2004)
## The Peter Principle: A Theory of Decline

# Contents

# 1 Big picture

- **Question.** Why do individuals often appear to perform worse after promotion, and does this imply that firms systematically make mistaken promotion decisions?
- **Core answer.** The observed post-promotion decline follows from regression to the mean. Promotion is awarded after a worker exceeds a standard. The observed performance that clears the standard contains permanent ability plus a favorable transitory component. The next-period transitory component has mean zero, so expected observed performance falls after promotion.
- **Main reinterpretation of the Peter principle.** The Peter principle does not require organizational irrationality. Even a firm using the correct promotion rule will observe promoted workers doing worse after promotion than before promotion.
- **Firm response.** Firms understand regression to the mean and adjust promotion standards. In the usual case, when fewer than half of workers should be promoted, the firm raises the promotion cutoff above the static job-switch point.
- **Key parameter.** The severity of the Peter principle depends on the size of the transitory component relative to the permanent component. More luck or measurement noise creates stronger regression and therefore a larger required adjustment to the promotion standard.
- **Job-specific interpretation.** If the transitory component is interpreted as job-specific skill, a worker can be excellent in the old job yet only average or below expectation in the new job. This gives a formal version of being promoted to one's level of incompetence.
- **Strategic effort extension.** If workers know their ability and firms assign jobs based on noisy output, workers may manipulate effort before the promotion decision. High-ability workers may overwork to increase promotion probability; low-ability workers may underwork to avoid an undesirable promotion.
- **Probation-period result.** When the transitory component is large, firms optimally wait longer before promotion because more observations improve the signal about permanent ability. When transitory noise is small, earlier promotion is optimal.
- **Broader logic.** The same statistical logic explains why sequels are often worse than originals, why second restaurant visits are less satisfying than exceptional first visits, why athletes may decline after appearing on magazine covers, and why reelected politicians may look worse in their second terms absent learning effects.
- **Takeaway.** Decline after selection is not necessarily evidence of error. It is an equilibrium implication of selecting on noisy performance measures.

# 2 Incremental contribution of the paper

## 2.1 Contribution relative to the informal Peter principle

- The original Peter principle claims that people are promoted to their level of incompetence and often treats this as a pathology of organizations.
- Lazear's contribution is to give a simple statistical and economic model in which the Peter principle arises even when firms are rational and promotion rules are optimal.
- The model separates two distinct ideas: (i) workers perform worse after promotion relative to their pre-promotion observed performance; and (ii) the promotion decision was a mistake. The paper shows that (i) can be true while (ii) is false.

**Interpretation:** The paper transforms the Peter principle from a management slogan into an economic selection problem under uncertainty.

### 2.2 Contribution relative to personnel economics and promotion theory

- Earlier personnel economics emphasized tournaments, incentives, wage dynamics, or job assignment.
- Lazear emphasizes an omitted statistical force: noisy signals used for promotion mechanically generate regression to the mean.
- The paper complements tournament models by showing that post-promotion decline can arise even without effort incentives and moral hazard.
- The paper also clarifies how tournaments and statistical regression differ: in tournament models, both winners and losers may reduce effort after the contest; in the pure regression model, winners' performance falls but losers' performance rises.

**Interpretation:** The paper's incremental contribution is a clean benchmark: even absent incentive distortions, promotion systems generate apparent performance decline.

### 2.3 Contribution relative to learning and job assignment models

- In learning models, firms gradually infer worker type and assign workers to tasks accordingly.
- Lazear adds the implication that selection on noisy signals causes post-selection reversal, even when learning and assignment are optimal.
- The paper also provides a criterion for when firms should delay promotion: longer probation is more valuable when transitory noise is large.

**Interpretation:** The article links promotion timing, signal quality, and assignment efficiency in a very compact way.

### 2.4 Contribution beyond labor economics

- The model applies to many selection decisions beyond promotions: sequels, repeat restaurant visits, political reelections, and media recognition.
- The general principle is: whenever continuation is conditional on an unusually good first draw, the second draw tends to look worse.
- This gives an economic analogue of the winner's curse, but with emphasis on repeated outcomes over time rather than bidding against others.

**Interpretation:** The paper's broader contribution is to show that apparent disappointment after success is a general selection phenomenon, not an idiosyncratic management failure.

## 3 Data, evidence, and motivating empirical facts

### 3.1 Why the topic matters empirically

The paper is primarily theoretical, but it is motivated by several empirical regularities in personnel economics and organizational behavior. Prior evidence suggests that workers' performance ratings, wage growth, or compensation increases can decline with time in a job. Lazear interprets these findings as consistent with selective promotion out of lower ranks and regression to the mean among promoted workers.

- Medoff and Abraham report that subjective evaluations fall with time on a job.
- Lazear's earlier work finds that, conditional on tenure in the firm, longer time in a specific job is associated with lower wages.

- Baker, Gibbs, and Holmstrom replicate related internal labor market patterns.
- Gibbs and Hendricks find that raises and bonuses decline with tenure.

**Interpretation:** The empirical facts do not by themselves prove mistakes. They are also compatible with optimal sorting and selection on noisy signals.

### 3.2 What the model is designed to explain

The model explains three observed patterns:

1. Promoted workers appear less able after promotion than before promotion.
2. Firms rarely reverse promotions, even though post-promotion performance falls.
3. Longer probation periods are more useful when performance contains a large temporary or noisy component.

**Interpretation:** The theory targets the direction of performance change, the optimality of the promotion rule, and the timing of promotions.

## 4 Baseline model

### 4.1 Ability components

Each worker has permanent ability $A$ and transitory components $e_1$ and $e_2$ in periods 1 and 2. The firm observes

$$\widehat{A} = A + e_1 \tag{1}$$

in period 1, but cannot separately observe $A$ and $e_1$.

- $A \sim f(A)$ is time-invariant ability.
- $e_t \sim g(e)$ is independently and identically distributed transitory ability or measurement error.
- $E[e_t] = 0$.

**Interpretation:** The central friction is signal contamination. The promotion signal combines lasting talent and temporary luck.

### 4.2 Two jobs

There are two jobs, an easy job and a difficult job. Productivity in the easy job is

$$\alpha + \beta(A + e_t), \tag{2}$$

while productivity in the difficult job is

$$\gamma + \delta(A + e_t), \tag{3}$$

with $\alpha > \gamma$ and $\delta > \beta$.

The difficult job has a lower intercept but a steeper slope in ability. It is efficient to assign a worker to the difficult job when

$$A + e_t > x, \tag{4}$$

where

$$x = \frac{\alpha - \gamma}{\delta - \beta}. \tag{5}$$

**Interpretation:** High-ability workers have comparative advantage in the difficult job. The job assignment decision is therefore a threshold rule.

### 4.3 Promotion rule

The firm chooses a promotion threshold $A^*$ and promotes if

$$\widehat{A} = A + e_1 > A^*. \tag{6}$$

The firm's period-2 assignment problem is to choose $A^*$ to maximize expected output:

$$\max_{A^*} \int_{-\infty}^{\infty} \int_{A^*-e_1}^{\infty} (\gamma + \delta A) dF dG + \int_{-\infty}^{\infty} \int_{-\infty}^{A^*-e_1} (\alpha + \beta A) dF dG. \tag{7}$$

**Interpretation:** The threshold is chosen using expected permanent ability, not observed first-period ability. That is why the optimal cutoff generally differs from the static crossing point $x$.

## 5 Main statistical result: post-promotion decline

### 5.1 Positive transitory component before promotion

For a promoted worker,

$$E(e_1 \mid A + e_1 > A^*) > 0. \tag{8}$$

This is because promotion selects workers with high observed ability. High observed ability comes partly from high permanent ability and partly from a favorable period-1 transitory component.

### 5.2 Zero transitory component after promotion

Because $e_2$ is independent of $A$ and $e_1$,

$$E(e_2 \mid A + e_1 > A^*) = 0. \tag{9}$$

Therefore,

$$E(A + e_2 \mid A + e_1 > A^*) < E(A + e_1 \mid A + e_1 > A^*). \tag{10}$$

**Interpretation:** This is the Peter principle in statistical form. Promoted workers look worse after promotion because the luck component that helped them cross the promotion threshold disappears.

### 5.3 Why this is not a mistake

The firm chooses $A^*$ knowing that promoted workers' measured performance will regress downward. If the firm has chosen $A^*$ optimally, expected output in the difficult job is still high enough to justify promotion.

- Decline relative to pre-promotion observed performance does not mean decline below the efficient assignment threshold.
- The firm should not usually undo the promotion.
- The rarity of demotions is more consistent with an optimal-threshold view than with the view that promotions are frequent mistakes.

**Interpretation:** A worker can do worse after promotion and still be the right person to promote.

# 6 Optimal promotion threshold

## 6.1 Why the threshold is usually inflated

If fewer than half of workers should be promoted, false positives are typically more costly than false negatives. The firm therefore raises the observed-ability threshold above the static switching point:

$$A^* > x. \tag{11}$$

**Details:** Lazear shows this formally under symmetric, unimodal distributions. Intuitively, a very high observed score is needed to compensate for the fact that some of the score reflects transitory luck.

## 6.2 Numerical examples

The paper considers the case where $A$, $e_1$, and $e_2$ are normally distributed with mean zero and variance one, and where

$$\alpha = 1, \quad \beta = 0.5, \quad \gamma = 0, \quad \delta = 1. \tag{12}$$

Then the static crossing point is

$$x = 2, \tag{13}$$

but the optimal promotion threshold is

$$A^* = 4.01. \tag{14}$$

If the standard deviation of $e_1$ falls from one to 0.1, the optimal cutoff falls to

$$A^* = 2.08. \tag{15}$$

**Interpretation:** As performance signals become less noisy, the promotion standard moves closer to the true job-switch point. When there is no noise, $A^* = x$.

# 7 Job-specific interpretation

## 7.1 Skill fit across jobs

The transitory component can also be interpreted as job-specific ability rather than pure luck. Let $e_1$ be the component relevant to the easy job and $e_2$ the component relevant to the difficult job.

- A worker may be excellent in the easy job because $e_1$ is high.
- That same worker may have average job-specific ability in the difficult job because $e_2$ is independent of $e_1$.
- The worker can therefore be promoted based on a skill that does not fully transfer.

**Interpretation:** This gives a formal version of the common statement that people promoted out of one job may not excel in the next one.

## 7.2 Being stuck at a job

Workers not promoted out of a job tend to have lower job-specific ability relative to workers entering that job. Those who are good are promoted out; those who remain have lower expected job-specific components.

$$E(e_1 \mid A + e_1 \leq A^*) < 0. \tag{16}$$

**Interpretation:** In a hierarchy with many steps, workers keep being promoted until they reach a level where they are no longer above the relevant threshold. This is the Peter principle in a multilevel hierarchy.

# 8 Strategic behavior by workers

## 8.1 Efficient effort when workers choose jobs

If workers are paid piece rates and can choose their jobs, they internalize the productivity consequences of their job choices. In that case, effort and job choice can be efficient.

The worker chooses the difficult job if

$$\gamma + \delta(A + m_d) - C(m_d) > \alpha + \beta(A + m_e) - C(m_e). \tag{17}$$

**Interpretation:** When workers choose both effort and jobs and are paid output-based piece rates, no promotion manipulation is necessary.

## 8.2 Tournaments and pre-promotion effort

If second-period wages depend only on promotion, workers compete against a promotion standard. The worker's first-period effort solves a tournament-like problem:

$$\max_{m_1} \int [W_d \Pr(A + m_1 + e_1 > A^*) + W_e(1 - \Pr(A + m_1 + e_1 > A^*)) - C(m_1)]\, dG(A). \tag{18}$$

The first-order condition implies that effort rises with the wage spread:

$$(W_d - W_e)g(A^* - m_1 - A) = C'(m_1). \tag{19}$$

**Interpretation:** Tournaments generate a separate reason for output to fall after promotion: effort incentives are weaker once the tournament is over.

## 8.3 Distortions when firms assign jobs

In the appendix, Lazear studies the case in which workers know their ability but firms assign jobs based on observed performance. High-ability workers may overwork to increase the probability of promotion, while low-ability workers may underwork to avoid being promoted into the difficult job.

**Interpretation:** Strategic effort can amplify or complicate the Peter principle, but it is not needed for the basic regression-to-the-mean result.

# 9 Probation length

## 9.1 Trade-off in delaying promotion

Longer probation improves information about permanent ability but delays correct assignment of high-ability workers to the difficult job.

- Waiting gives more accurate information.
- Waiting keeps some able workers in the easy job too long.
- Early promotion sorts workers sooner but makes more mistakes.

**Interpretation:** The promotion timing problem is an information-versus-allocation trade-off.

## 9.2 Comparing early versus late decisions

If waiting two periods gives a perfect reading of $A$, expected lifetime output under waiting is

$$2\alpha + \int_{x}^{\infty} (\gamma + \delta A) dF + \int_{-\infty}^{x} (\alpha + \beta A) dF. \tag{20}$$

If the firm decides early using the noisy signal, expected output is based on the noisy threshold $A^*$ and therefore contains more assignment mistakes.

**Interpretation:** When transitory noise is small, early promotion dominates. When transitory noise is large, delayed promotion can be optimal.

## 9.3 Main comparative static

As the variance of the transitory component rises, the value of waiting rises:

$$\mathrm{Var}(e) \uparrow \quad \Rightarrow \quad \text{longer probation becomes more attractive.} \tag{21}$$

**Interpretation:** High-noise jobs should have longer evaluation periods before promotion.

# 10 Broader applications

## 10.1 Movie sequels

A movie sequel is made only when the original movie is sufficiently successful. Let the original movie's performance be

$$A + e_1, \tag{22}$$

where $A$ is the durable movie concept or franchise component and $e_1$ is the temporary realization of execution, casting, timing, or luck. A sequel is made only if

$$A + e_1 > A^*. \tag{23}$$

Since

$$E(e_1 \mid sequel) > 0, \tag{24}$$

but

$$E(e_2 \mid sequel) = 0, \tag{25}$$

the expected sequel is worse than the original.

**Interpretation:** Poorer sequel performance is not necessarily evidence that studios make mistakes. It is selection on unusually successful originals.

## 10.2 Restaurants and repeat experiences

The same logic applies to restaurants. A person returns after a very good meal, but that meal contains both a permanent restaurant component and a transitory component. The next meal regresses toward the permanent mean.

**Interpretation:** The first exceptional visit overstates the expected quality of the next visit.

### 10.3 Sports and politics

Athletes chosen for magazine covers and politicians chosen for reelection are selected after high observed performance. Unless learning or incumbency advantages are strong, future performance should regress downward.

**Interpretation:** Post-selection disappointment is not itself proof of bad selection.

## 11 Figures

The paper contains two visual figures. They are placed on the same row, following the requested compact layout.

**Read:**

- Figure 1 shows the productivity schedules for easy and difficult jobs. The difficult job has lower intercept but higher slope. The static switch point is $x$, while the optimal promotion cutoff $A^*$ is typically above $x$ when firms adjust for regression to the mean.
- Figure A1 shows effort distortion when firms assign jobs but workers know their abilities. Workers far from the promotion margin do not distort much. Workers below the switch point may underwork to avoid promotion; workers above the switch point may overwork to obtain promotion.

**Economic message:** The two figures summarize the paper's two main mechanisms: promotion thresholds under noisy performance and strategic effort distortions around the promotion margin.

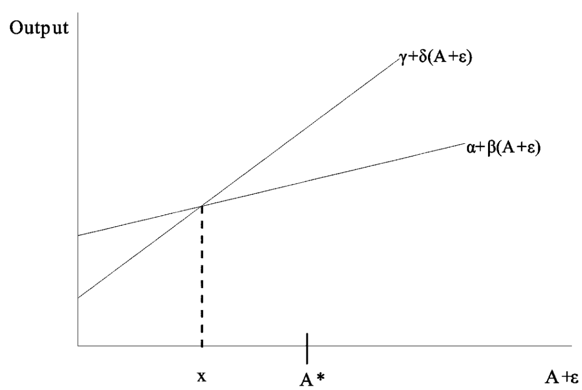


Fig. 1

*Figure 1: easy and difficult job output schedules*

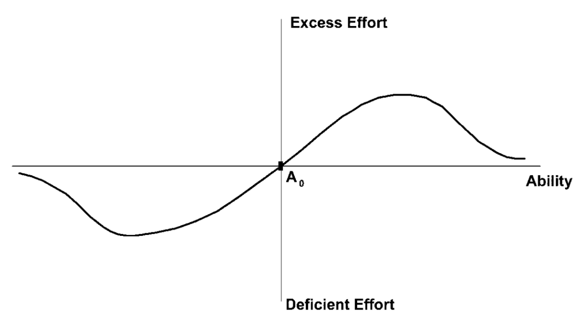


Fig. A1

*Figure A1: effort distortions around the promotion threshold*

## 12 Short summary

- The Peter principle can arise even when firms make optimal promotion decisions.
- Promotion is based on noisy observed performance $A + e_1$.
- Conditional on promotion, the first-period transitory component is positive.
- The next-period transitory component has expectation zero.
- Therefore promoted workers look worse after promotion than before promotion.
- Firms anticipate this and usually raise promotion cutoffs above the static productivity crossing point.

- The adjustment is larger when transitory variation is larger.
- Longer probation periods are more valuable when transitory noise is high.
- If transitory ability is job-specific, workers can be promoted from jobs where they fit well into jobs where they fit less well.
- Strategic effort can distort pre-promotion performance when workers know more about their ability than firms do.
- The same logic applies to sequels, restaurants, sports recognition, and political reelection.

# 13 Conclusion

## 13.1 Core conclusion

Lazear shows that the Peter principle does not require irrational firms, biased managers, or defective hierarchies. It follows from selecting workers for promotion on the basis of noisy performance. The promoted group had unusually good first-period observations partly because of luck or job-specific fit, and those favorable transitory components do not persist.

## 13.2 Economic intuition

The firm faces a selection problem. It wants to promote workers whose permanent ability makes them suitable for a more difficult job, but it observes only a noisy signal. Because promotion selects high signals, promoted workers' pre-promotion performance is biased upward. Rational firms raise promotion standards to compensate, but they cannot eliminate regression to the mean.

## 13.3 Takeaway

The paper's main lesson is that decline after success is a statistical consequence of selection. The correct policy response is not necessarily to reverse promotions or abandon hierarchies. It is to account for signal noise, choose promotion standards carefully, and lengthen probation when performance contains a large transitory component.